

Sharp heat smoothing for sign-aligned Walsh spectra

Autonomous solution packet for arXiv:1406.7855

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Status: candidate partial result, likely valid. For every even p and for $p = \infty$, the full heat-smoothing estimate holds with sharp constant one on a translation-invariant cone of tail functions. The unrestricted scalar conjecture remains open.

1 The conjecture and the coefficient cone

On the discrete cube with uniform probability measure, write

$$f = \sum_{S \subseteq [n]} \widehat{f}(S) W_S, \quad W_S(x) = \prod_{i \in S} x_i,$$

and let

$$P_t f = \sum_S e^{-t|S|} \widehat{f}(S) W_S, \quad Lf = \sum_S |S| \widehat{f}(S) W_S.$$

The k -th tail space consists of functions for which $\widehat{f}(S) = 0$ whenever $|S| < k$. The Mendel–Naor heat-smoothing conjecture asks whether, for every $1 < p < \infty$, there is $c_p > 0$ such that

$$\|P_t f\|_p \leq e^{-c_p k t} \|f\|_p \tag{HS}$$

for every dimension, tail degree, and scalar tail function.

A real function f has a *sign-aligned Walsh spectrum* if there exist $\varepsilon \in \{-1, 1\}$ and $\tau \in \{-1, 1\}^n$ such that

$$\varepsilon W_S(\tau) \widehat{f}(S) \geq 0 \quad (S \subseteq [n]). \tag{SA}$$

The condition is invariant under cube translations and overall sign. It imposes no upper bound on the Fourier degrees and therefore differs from the narrow-spectrum hypotheses treated in later work.

Theorem 1 (sharp even-moment heat smoothing). *Let $k \geq 1$, let $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ belong to the k -th tail space, and suppose that f satisfies (SA). Then, for every integer $m \geq 1$ and every $t \geq 0$,*

$$\|P_t f\|_{2m} \leq e^{-k t} \|f\|_{2m}. \tag{1}$$

The same estimate holds for $p = \infty$. The rate is sharp.

2 The zero-symmetric-difference expansion

The proof rests on the following elementary lemma.

Theorem 2 (positive-coefficient coercivity). *Suppose*

$$g = \sum_{\substack{S \subseteq [n] \\ |S| \geq k}} b_S W_S, \quad b_S \geq 0.$$

Then, for every integer $m \geq 1$,

$$\mathbb{E}[g^{2m-1} Lg] \geq k \mathbb{E}[g^{2m}]. \quad (2)$$

Proof. Orthogonality of the Walsh characters, together with $W_{S_1} \cdots W_{S_r} = W_{S_1 \Delta \cdots \Delta S_r}$, gives

$$\mathbb{E}[g^{2m}] = \sum_{\substack{S_1, \dots, S_{2m} \\ S_1 \Delta \cdots \Delta S_{2m} = \emptyset}} \prod_{j=1}^{2m} b_{S_j}. \quad (3)$$

Similarly,

$$\mathbb{E}[g^{2m-1} Lg] = \sum_{\substack{S_1, \dots, S_{2m} \\ S_1 \Delta \cdots \Delta S_{2m} = \emptyset}} |S_{2m}| \prod_{j=1}^{2m} b_{S_j}. \quad (4)$$

The index set and coefficient product in (4) are invariant under every permutation of the $2m$ positions. Averaging the distinguished degree over all positions therefore rewrites (4) as

$$\frac{1}{2m} \sum_{\substack{S_1, \dots, S_{2m} \\ S_1 \Delta \cdots \Delta S_{2m} = \emptyset}} \left(\sum_{j=1}^{2m} |S_j| \right) \prod_{j=1}^{2m} b_{S_j}. \quad (5)$$

Every nonzero product in (5) has $|S_j| \geq k$ for all j , and every product is nonnegative. Thus (5) is at least k times (3). $\square$

3 Proof of Theorem 1

Proof. Translations $f(x) \mapsto f(\tau x)$ and multiplication by an overall sign preserve every L_p norm and commute with P_t and L . We may therefore assume $\hat{f}(S) \geq 0$ for every S .

Put $g_t = P_t f$. Its Walsh coefficients $e^{-t|S|} \hat{f}(S)$ remain nonnegative and remain supported on degrees at least k . Differentiating the finite sum and applying Theorem 2 gives

$$\begin{aligned} \frac{d}{dt} \mathbb{E}[g_t^{2m}] &= -2m \mathbb{E}[g_t^{2m-1} Lg_t] \\ &\leq -2mk \mathbb{E}[g_t^{2m}]. \end{aligned}$$

Gronwall's inequality yields $\mathbb{E}[(P_t f)^{2m}] \leq e^{-2mkt} \mathbb{E}[f^{2m}]$, which is (1).

For the endpoint, nonnegativity of the coefficients shows

$$\|f\|_\infty = \sum_S \hat{f}(S) :$$

the triangle inequality gives the upper bound and equality is attained at the all-ones point. Consequently

$$\|P_t f\|_\infty = \sum_S e^{-t|S|} \hat{f}(S) \leq e^{-kt} \sum_S \hat{f}(S) = e^{-kt} \|f\|_\infty.$$

Finally, a single Walsh character W_S with $|S| = k$ gives equality for every p , proving sharpness. $\square$

4 Scope and obstruction to a full result

For arbitrary real coefficients, formulas (3)–(5) remain true but their products can have both signs. The termwise degree comparison then fails: the general conjecture is exactly asking for a way to control those cancellations in a nonlinear L_p pairing.

Replacing the coefficients by their absolute values produces a sign-aligned majorant, but the Walsh basis has no applicable dimension-free unconditional estimate that compares its norm with the original function. Interpolation between adjacent even moments also has the initial norms in the wrong direction. Thus this argument does not resolve non-even p or unrestricted coefficient signs.

The source paper proves the degree-one case for all p and the arbitrary-degree case for three-valued functions. Subsequent work gives narrow-spectrum and holomorphic special cases, while still describing the unrestricted scalar cube conjecture as open. Targeted searches found no matching sign-aligned even-moment theorem, but the novelty search was not exhaustive.

The accompanying script `code/verify_even_moment.py` enumerates a small cube and checks (2) for random nonnegative coefficients at $p = 4$. It is a numerical sanity check only.

References

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- [4] H. Zhang, *Heat-smoothing for holomorphic subalgebras of free group von Neumann algebras*, arXiv:2205.12833.

Source context

Page 1 of the locally rendered arXiv:1406.7855 source states the full heat-smoothing conjecture. The abstract also records which influence questions are answered in the same paper.

STRONG CONTRACTION AND INFLUENCES IN TAIL SPACES

STEVEN HEILMAN, ELCHANAN MOSSEL, AND KRZYSZTOF OLESZKIEWICZ

ABSTRACT. We study contraction under a Markov semi-group and influence bounds for functions in L^2 tail spaces, i.e. functions all of whose low level Fourier coefficients vanish. It is natural to expect that certain analytic inequalities are stronger for such functions than for general functions in L^2 . In the positive direction we prove an L^p Poincaré inequality and moment decay estimates for mean 0 functions and for all $1 < p < \infty$, proving the degree one case of a conjecture of Mendel and Naor as well as the general degree case of the conjecture when restricted to Boolean functions. In the negative direction, we answer negatively two questions of Hatami and Kalai concerning extensions of the Kahn-Kalai-Linial and Harper Theorems to tail spaces. That is, we construct a function $f: \{-1, 1\}^n \rightarrow \{-1, 1\}$ whose Fourier coefficients vanish up to level $c \log n$, with all influences bounded by $C \log n/n$ for some constants $0 < c, C < \infty$. We also construct a function $f: \{-1, 1\}^n \rightarrow \{0, 1\}$ with nonzero mean whose remaining Fourier coefficients vanish up to level $c' \log n$, with the sum of the influences bounded by $C'(\mathbb{E}f) \log(1/\mathbb{E}f)$ for some constants $0 < c', C' < \infty$.

1. INTRODUCTION

Consider the uniform measure on $\{-1, 1\}^n$. Any $f: \{-1, 1\}^n \rightarrow \mathbb{R}$ can be written as $f = \sum_{S \subseteq \{1, \dots, n\}} \hat{f}(S) W_S$, where for all $x = (x_1, \dots, x_n) \in \{-1, 1\}^n$, $W_S(x) := \prod_{i \in S} x_i$ and $\hat{f}(S) := 2^{-n} \sum_{x \in \{-1, 1\}^n} f(x) W_S(x)$. For any $t \geq 0$, define $P_t f := \sum_{S \subseteq \{1, \dots, n\}} e^{-t|S|} \hat{f}(S) W_S$, and define $Lf := \sum_{S \subseteq \{1, \dots, n\}} |S| \hat{f}(S) W_S$.

Our interest in this paper is in tail spaces. For the case of the uniform measure on $\{-1, 1\}^n$, we are interested in the linear subspace of all functions satisfying $\hat{f}(S) = 0$ for all $S \subseteq \{1, \dots, n\}$ with $|S| \leq k$. Our interest in understanding such functions follows recent conjectures by Mendel and Naor and by Hatami and Kalai.

1.1. Heat Smoothing. In their study of a general notion of expander (with respect to all uniformly convex spaces), Mendel and Naor made the following conjecture:

Conjecture 1.1 (Heat Smoothing). [13] Remark 5.5] *Let $1 < p < \infty$. Let $f: \{-1, 1\}^n \rightarrow \mathbb{R}$ with $\mathbb{E}f W_S = 0$ for all $S \subseteq \{1, \dots, n\}$ with $|S| < k$. Then*

$$\forall t > 0, \quad \|P_t f\|_p \leq e^{-tkc(p)} \|f\|_p. \quad (1.1)$$

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